



# Incidence and Molecular Characterization of Carbapenemase Genes in Association with Multidrug-Resistant Clinical Isolates of *Pseudomonas aeruginosa* from Tertiary Healthcare Facilities in Southwest Nigeria

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## Abstract

*Pseudomonas aeruginosa*, resistant to multiple antibacterial agents including carbapenems, is of great global public health concern. There is limited data available regarding incidence of Metallo-Beta Lactamase producing *P. aeruginosa*, their molecular basis of resistance in particular carbapenem resistance and any genetic relatedness among circulating clinical isolates in Southwest Nigeria. Four hundred and thirty *P. aeruginosa* isolates were collected from seven tertiary care hospitals (predominantly from wound, ear, and urinary tract infections) and verified by PCR targeting *oprI* and *oprL*. Antibiotic susceptibility using 16 selected antibiotics and MBL screening was performed. The integrons (class 1, 2 and 3) and carbapenemase genes- *bla*<sub>GES</sub>, *bla*<sub>NMC-A</sub>, *bla*<sub>BIC-1</sub>, *bla*<sub>SME</sub>, *bla*<sub>IMP</sub>, *bla*<sub>VIM</sub>, *bla*<sub>SPM</sub>, *bla*<sub>NDM</sub>, *bla*<sub>AIM</sub>, *bla*<sub>DIM</sub>, *bla*<sub>SIM</sub>, *bla*<sub>GIM</sub>, *bla*<sub>OXA-48</sub>, *bla*<sub>OXA-58</sub> were detected by PCR and were sequenced. Quantitative real-time polymerase chain reaction was used to quantify expression levels of eight efflux pump genes, *ampC* cephalosporinase and outer membrane porin, *oprD*. The isolates were genotyped using Enterobacterial Repetitive Intergenic Consensus sequence Polymerase Chain Reaction (ERIC-PCR). Four hundred and thirty *P. aeruginosa* isolates were subjected to antibiotic susceptibility testing, revealing that 109 (25.4%) isolates were multidrug-resistant, 47 (10.9%) were extensively drug-resistant and 25 (5.8%) were pandrug-resistant. MBL was seen in 17.0% (73/430) isolates. MBL-encoding genes; *bla*<sub>VIM-5</sub> and *bla*<sub>NDM-1</sub> were detected in 86.3% (63/73) isolates, with *bla*<sub>VIM-5</sub> and *bla*<sub>NDM-1</sub> in 35.6% (26/73) and 38.4% (28/73), respectively, whereas co-occurrence of *bla*<sub>VIM-5</sub> and *bla*<sub>NDM-1</sub> was found in 12.3% (9/73). Forty-one (56.2%) carbapenem-resistant *P. aeruginosa* strains carried class 1 integrons, while co-occurrence of class 1 and 2 integrons was seen in 12.3%. qPCR results indicated that MexXY-OprM was highly expressed pump in 58.9%, *ampC* upregulated in 26.0%, while *oprD* porin was downregulated in 65.8% isolates. ERIC-PCR results suggest that carbapenem-resistant strains exhibit genetic heterogeneity. The high incidence of MBL-encoding genes and integrons in diversified clinical *P. aeruginosa* from southwestern Nigeria is of great concern. The co-occurrence of *bla*<sub>VIM-5</sub> and *bla*<sub>NDM-1</sub> as well as resistance in general manifesting a gradient based on genotypic variation suggests that there is a strong need for efficient surveillance programs and antibiotic stewardship.

## Introduction

Nosocomial infections with extremely drug-resistant Gram-negative organisms are increasingly becoming a significant problem of global public health concern. *Pseudomonas aeruginosa* is an opportunistic Gram-negative bacterium implicated in causing severe and fatal infections in immunocompromised hosts but rarely causes disease in healthy individuals [1]. *P. aeruginosa* infections have been acknowledged as major global health challenge mainly because of inherent resistance of this bacterium to many classes of antibiotics besides its ability to acquire resistance to virtually

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all potent antibiotics [2]. It is the most common nosocomial pathogen reported from most hospital based studies followed by *Staphylococcus aureus* and is an extremely important member of ESKAPE group which escape the bactericidal action of antibiotics [3, 4]. This ESKAPE group includes six significant nosocomial pathogens consisting of *Enterococcus faecium*, *Staphylococcus aureus*, *Klebsiella pneumoniae*, *Acinetobacter baumannii*, *Pseudomonas aeruginosa* and *Enterobacter* species [3]. Multidrug-resistant *P. aeruginosa* was ranked among the first category of pathogens causing life-threatening infections and high priority pathogens that urgently require new antibiotics [5].

The introduction of carbapenems (broad-spectrum antibiotics) into clinical use signifies a pronounced progress in the treatment of multidrug-resistant organisms and is usually reserved for the treatment of serious Gram-negative bacterial infections [6]. However, resistance to carbapenems is on the rise and has rendered the clinical efficacy of existing panel of carbapenems irrelevant [6, 7]. Carbapenem resistance in *P. aeruginosa* is mediated by different mechanisms such as acquired carbapenemases, upregulation of efflux pump systems, overexpression of chromosomal cephalosporinase and decreased outer membrane permeability [5, 8]. However, acquired carbapenemases particularly the metallo-beta lactamases (MBLs) are the most clinically significant of all carbapenemases because of their broad hydrolysis profile [6]. MBL-producing organisms are known to be resistant to almost all beta-lactam antibiotics with the exception of aztreonam making polymyxins the only available treatment of choice [9]; hence, there is a great need for systematically monitoring of these pathogens to better control the emerging antibiotic resistance threat. MBL has been reported in various Gram-negative bacteria and has been associated with mobile genetic elements such as integrons and plasmids which aid in their rapid dissemination [10].

Recent study suggests that there are ten types of MBLs originally present, with further diverse variants within each type of MBL with blaIMP and blaVIM being the most predominant [11]. The presence of bla<sub>KPC</sub>, bla<sub>NDM</sub> and bla<sub>VIM</sub> type carbapenemase in carbapenem-resistant Enterobacteriaceae and bla<sub>VIM-2</sub> in *P. aeruginosa* from Northern Nigeria have been previously documented [12, 13]. The routine basal expression of efflux pumps in clinical isolates of *P. aeruginosa* play important part in conferring moderately reduced susceptibility to antibiotics but overexpression of these efflux pumps in natural variants or mutants lead to raised level of resistance to antibiotics [14, 15]. Efflux pumps have capacity to eject numerous categories of antibiotics from the cytoplasm and the periplasmic space, subsequently leading to cross-resistance to other classes of antibiotics because reasonable inhibitory concentration of antibiotic could not be achieved at the target site [14]. Out of the 12 resistance nodulation division (RND) type efflux pumps so

far identified, four are of clinical importance. These include MexAB-OprM, MexCD-OprJ, MexEF-OprN and MexXY-OprM [16]. MexXY-OprM exports fluoroquinolones, aminoglycosides, and cefepime, while MexAB-OprM exports beta-lactams, macrolides, fluoroquinolones, novobiocin, sulfonamides, trimethoprim, tetracycline, chloramphenicol and meropenem [15]. The relationship between antibiotic resistance, in particular resistance to carbapenems in *P. aeruginosa* isolates, and linking it with the underlying genetic basis in Southwest Nigeria has not been extensively studied. The proposed study aims to understand the linkages between prevalent isolates of *P. aeruginosa* in Southwest Nigeria, their susceptibility profile to antibiotics and repertoire of genes conferring resistance against carbapenems. In addition, we also investigated the expression of efflux pump-encoding genes and further linking them with the enhanced resistance phenotype of *P. aeruginosa*. Moreover, the genetic heterogeneity of circulating highly resistant clinical isolates of *P. aeruginosa* has also been investigated using rapid and economical ERIC-PCR approach in resource deficient Laboratory setting.

## Materials and Methods

### Study Design and Bacterial Isolates

This descriptive cross-sectional study was done between September 2014 and April 2018 in Pharmaceutical Microbiology laboratory, University of Ibadan, Nigeria and Microbiology and Public Health Laboratory, COMSATS University Islamabad (CUI) Pakistan. Four hundred and thirty *P. aeruginosa* isolates predominantly from wound, ear, and urinary tract infections were obtained between September 2014 and June 2017 from Clinical Microbiology laboratories of seven tertiary hospitals located in Southwestern Nigeria. These included 45 isolates from Obafemi Awolowo University Teaching Hospital Complex, Ile-Ife (OTHI), 90 isolates from University College Hospital, Ibadan (UCHI), 45 isolates from Ladoke Akintola University Teaching Hospital, Osogbo (LTHO), 45 isolates from Federal Medical Centre, Abeokuta (FMCA), 45 isolates from Olabisi Onabanjo University Teaching Hospital, Sagamu (OTHS), 90 isolates from Federal Medical Centre, Owo (FMCO) and 70 isolates from Federal Medical Centre, Ido-Ekiti (FMCI). The isolates were preserved on nutrient agar slants in the refrigerator at 4 °C until when needed. Repeat samples were excluded from the study.

### Identification of Clinical Isolates

Gram staining was done to identify the organism structurally. Colony morphology and pigment production was determined

on *Pseudomonas* Cetrimide agar (Oxoid, UK). The isolates were streaked on MacConkey and blood agar to test for lactose fermentation and hemolysis, respectively. Other tests performed include oxidase test using oxidase strip (Oxoid, UK), gelatin liquefaction, citrate, catalase, indole and growth at 42 °C [17]. Isolates were further confirmed by PCR amplification of *oprI* and *oprL* genes specific for the genus *Pseudomonas* and *P. aeruginosa*, respectively [18, 19]. A positive control with *P. aeruginosa* ATCC 27853 DNA and a negative control without DNA were included.

### Isolation of Genomic DNA

DNA was extracted from *P. aeruginosa* strains grown overnight in 5 mL LB broth with WizPrep™ purification kit according to Manufacturer's instruction. Extracted DNA was eluted with 100 µL of elution buffer and ran on 1.5% agarose and visualized with a UV transilluminator using a gel documentation and Analysis System (GenoSens 1560). Quantity and purity of DNA was checked with NanoDrop® (Titertek Berthold, Germany) and kept at – 20 °C prior to use.

### Antimicrobial Susceptibility Patterns

Antibiogram of 430 isolates of *P. aeruginosa* against 16 antibiotics was determined using Kirby Bauer disk diffusion method. The results were interpreted according to CLSI guidelines [20]. Isolates were classified as, PDR, XDR or MDR according to the criteria for classification of acquired resistance in *P. aeruginosa* [21]. PDR was defined as resistance to all the antimicrobial agents tested including ticarcillin, gentamicin, amikacin, ofloxacin, ciprofloxacin, levofloxacin ceftazidime, cefepime, cefoperazone, tobramycin, piperacillin, piperacillin–tazobactam, doripenem, meropenem, imipenem and aztreonam. MDR was defined as resistance to at least one agent in three or more antimicrobial classes, while XDR was defined as resistance to at least one agent in all but two antimicrobial classes. Combined disk method was used to detect the presence of MBLs in clinical isolates of *P. aeruginosa* as described by Yong et al [22].

### Minimum Inhibitory Concentration (MIC) by Broth Microdilution Method

MIC determination was done by broth microdilution method using meropenem powder (Hi Media) according to CLSI guidelines [20]. *P. aeruginosa* ATCC 27853 serve as quality control organism.

### Amplification and Sequencing of Carbapenemase Genes

A total 25 µL reaction mixture containing 12.5 µL WizPure™ PCR 2X Master (Wizbiosolutions, Korea South), 0.5 µL each of 10 µmol/L forward and reverse primers (Alpha DNA) genes, 1 µL of genomic DNA and 10.5 µL DNase/RNase-Free Distilled Water (Invitrogen) in a 0.2 mL PCR tube was used for all amplification reactions in a singleplex reaction. Primers specific *bla*<sub>GES</sub>, *bla*<sub>NMC-A</sub>, *bla*<sub>BIC-1</sub>, *bla*<sub>SME</sub>, *bla*<sub>IMP</sub>, *bla*<sub>VIM</sub>, *bla*<sub>SPM</sub>, *bla*<sub>NDM</sub>, *bla*<sub>AIM</sub>, *bla*<sub>DIM</sub>, *bla*<sub>SIM</sub>, *bla*<sub>GIM</sub>, *bla*<sub>OXA-48</sub> and *bla*<sub>OXA-58</sub> (Supplementary table 1.1) were used to amplify carbapenemase genes in carbapenem non-susceptible *P. aeruginosa* clinical isolates. The tubes were loaded into Thermal cycler (Applied Biosystems) and the PCR was performed as reported previously. All PCR products were analyzed on a 1.5% agarose after staining with 0.5 µg/mL ethidium bromide. The bands were visualized with a UV transilluminator using a gel documentation and Analysis System (GenoSens 1560). The panel of shortlisted eighteen extremely diversified isolates were used to characterize the repertoire of amplified resistance gene products after purifying with WizPrep Gel/PCR purification kit (Wizbiosolutions, Korea South), it was subjected to sequencing using both strands with BigDye Terminator v3.1 Cycle Sequencing Kit on ABI PRISM 3730xi Sequence analyzer (Applied Biosystems, USA). The resulting sequences were compared with the sequences in the NCBI GenBank database through the BLAST platform (<http://blast.ncbi.nlm.nih.gov/Blast.cgi>) and deposited in the GenBank.

### Characterization of Integron by PCR

Integrations were amplified by PCR using degenerate primers, hep35 and hep36 (Supplementary table 1.1), which hybridize to conserved regions of integron-encoded integrase genes *intI1*, *intI2*, and *intI3* in a Thermocycler (Applied Biosystems, USA). PCR products were analyzed to determine the class of the integron by digesting the products with *RsaI* (Thermo Scientific) restriction enzyme according to previously described protocol [23]. All PCR products were analyzed on a 1.5% agarose containing 0.5 µg/mL of ethidium bromide. The bands were visualized with a UV transilluminator using a gel documentation and Analysis System (GenoSens 1560). Strains that gave amplicon size of 491 bps were subjected to another PCR using hep58 and hep59 primers to detect class 1 gene cassette arrays [24].

## Quantification of Gene Expression

### RNA Extraction and Synthesis of First Strand Complementary DNA (cDNA)

Total RNA was extracted from carbapenem-resistant *P. aeruginosa* with PureLink™ total RNA purification kit (Invitrogen) and treated with DNase 1 (Thermo Scientific) following manufacturer's instructions to remove traces of contaminating DNA. Complementary DNA was synthesized with cDNA Synthesis Kit (Wizbiosolutions, Korea) according to the manufacturer's instruction using RNA template of 0.5–4 µL (1 µg). Synthesized cDNA were kept at -20 °C and used as template for quantitative real-time PCR.

### Expression Levels Quantification of Efflux Pumps, *ampC* and *oprD* Transcripts

The expression levels of two genes in each of the four major *P. aeruginosa* efflux pumps; (*mexA*, *mexB*, *mexC*, *mexD*, *mexE*, *mexF*, *mexX* and *mexY*), *ampC* and *oprD* genes were analyzed with Step One ® real-time reverse transcription-PCR (Applied Biosystems, USA). The primer sequences for the PCR amplification of cDNA are listed in Supplementary material 1.2. Real-time PCR was accomplished in a set of three per sample. A total capacity of 20 µL comprised of 10 µL of SYBR green/ROX qPCR master mix (Thermo Scientific), 0.5 µL (10 pmol) each of forward and reverse primers, 1 µL of the cDNA, 8 µL of PCR water was used. PCR conditions include initial denaturation temperature at 95 °C for 10 min, followed by 40 cycles of 20s at 95 °C, 30s at 66 °C (*mexA*, *mexB*, *mexC*, *mexX* and *mexY*), 30s at 67 °C (*mexE*), 30s at 68 °C (*mexF*), 30s at 64 °C (*mexD*), 30s at 62 °C (*ampC*), 30s at 62 °C (*oprD*), 30s at 62 °C (*rpsL*) and 30s at 72 °C. Data acquisition was performed at 72 °C. Melt curve generated at the end of the 40 cycles was used to determine the absence of primer dimer. The mean threshold cycle (ct) value of triplicate sample was taken. Housekeeping gene (*rpsL* gene) was referenced against *P. aeruginosa* ATCC 27853 expressions to regulate the expression levels of mRNA. Fold expression levels was calculated using  $2^{-\Delta\Delta Ct}$ . Control reaction lacking reverse transcriptase was set up to check for contaminating genomic DNA. When the equivalent mRNA level was at least 2-fold (*mexA*, *mexB*), 4-fold (*mexX* and *mexY*) or 100-fold (*mexC*, *mexD*, *mexE* and *mexF*) higher than that for *P. aeruginosa* ATCC 27853, strains were considered to be overexpressed [25].

### ERIC-PCR-Based Genotyping

ERIC-1R 5'ATGTAAGCTCCTGGGGATTCAC3' and ERIC-2 5'AAGTAAGTGACTGGGTGAGCG3' primers were used [26]. ERIC-PCR was carried out according to

protocol of [27]. PCR set up included 1 µL of DNA template, 0.5 µL of each primer (Macrogen), 12.5 µL of Wiz-Pure™ PCR 2X Master mix (Wizbiosolutions, Korea South), and 7.5 µL of distilled water devoid of DNase and RNase (Invitrogen). PCR products were analyzed on a 1.5% agarose containing 0.5 µg/mL of ethidium bromide. The bands were visualized with a UV transilluminator using a gel documentation and Analysis System (GenoSens 1560). The arrangement of amplified DNA fragments was scored alongside with a DNA molecular weight marker (1 kb plus ladder). Data matrices were computed based on the presence or absence of bands at certain positions. ERIC-PCR profiles were compared with dice coefficient method and clustered with Unweighted Pair Group Method with Arithmetic Mean (UPGMA) using Phylotree software (<http://phylotree.hyphy.org/>).

### Statistical Analysis

Data were analyzed using Graphpad Prism 6 Software. Fisher's Exact test was used to analyze categorical data. The probability (*P*) value of ≤0.05 was considered statistically significant.

## Results

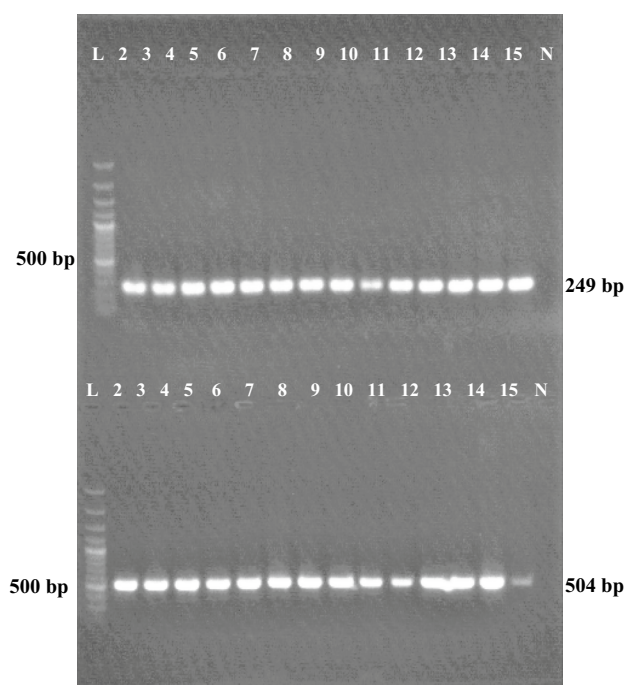
### Isolation, Demographic Characteristics and Distribution of *P. aeruginosa*

A total of 430 *P. aeruginosa* isolates were biochemically identified. The isolates were further confirmed by PCR amplification of *oprI* and *oprL* genes specific for the *P. aeruginosa* (Fig. 1). *P. aeruginosa* isolates was highest (56.5%) among male patients. Highest percentage (37.4%) of *P. aeruginosa* was obtained in age group 41–60 years followed by age group 21–40 years (33.0%) (Supplementary table 2). The isolates were obtained from twelve clinical sources including wound (298), ear (49), urine (43), wound biopsy (5), pus (5), sputum (5), eye (10), catheter tips (4), blood (4), cerebrospinal fluid (3), tracheal aspirate (3) and genital discharge (1). Detailed information on the occurrence of *P. aeruginosa* strains according to hospital with respect to their clinical sources is shown in Supplementary table 3.

### Antibiotic Resistance Profiling of *P. aeruginosa* and Phenotypic MBL Detection

Table 1 shows the antibiotic susceptibility profile of the 430 clinical isolates. The resistance rates in descending order were: ticarcillin (64.4%), cefepime (45.8%), gentamicin (43.3%), ofloxacin (43.3%), ciprofloxacin (39.3%), ceftazidime (38.1%), levofloxacin (38.1%), cefoperazone (33.0%),





**Fig. 1** Row 1: Agarose gel electrophoresis of PCR products (1.5%) for the identification of *Pseudomonas* spp. using *oprI* Genus specific primers L: 100 bp plus ladder; Lane 2 ATCC 27,853; Lane 3–15 (PS296, PS325, PS97, PS100, PS185, PS182, PS154, PS168, PS173, PS204, PS210, PS224, PS414); N=*Escherichia coli* (ATCC 25,922). Row 2: PCR products for the identification of *P. aeruginosa* using *oprL* species specific primers L: 100 bp plus ladder; Lane 2 ATCC 27,853; Lane 3–15 (PS296, PS325, PS97, PS100, PS185, PS182, PS154, PS168, PS173, PS204, PS210, PS224, PS414), N=*Escherichia coli* (ATCC 25,922)

**Table 1** Antibiotic susceptibility profile of clinical isolates of *P. aeruginosa*

| Antibiotics              | Sensitive (%) | Intermediate (%) | Resistance (%) |
|--------------------------|---------------|------------------|----------------|
| Ceftazidime              | 214 (49.8)    | 47 (10.9)        | 164 (38.1)     |
| Piperacillin             | 299 (69.5)    | 51 (11.9)        | 80 (18.6)      |
| Ticarcillin              | 32 (7.4)      | 121 (28.1)       | 277 (64.4)     |
| Cefoperazone             | 211 (49.1)    | 77 (17.9)        | 142 (33.0)     |
| Piperacillin/ tazobactam | 303 (70.5)    | 51 (11.9)        | 76 (17.7)      |
| Gentamicin               | 214 (49.8)    | 30 (7.0)         | 186 (43.3)     |
| Imipenem                 | 360 (83.7)    | 2 (0.5)          | 68 (15.8)      |
| Aztreonam                | 212 (49.3)    | 146 (34.0)       | 72 (16.7)      |
| Tobramycin               | 284 (66.1)    | 6 (1.4)          | 140 (32.6)     |
| Meropenem                | 351 (81.6)    | 8 (1.9)          | 71 (16.5)      |
| Amikacin                 | 29 (69.5)     | 13 (3.0)         | 118 (27.4)     |
| Levofloxacin             | 259 (60.2)    | 7 (1.6)          | 164 (38.1)     |
| Doripenem                | 349 (80.2)    | 8 (1.9)          | 73 (17.0)      |
| Cefepime                 | 203 (47.2)    | 30 (7.0)         | 197 (45.8)     |
| Ciprofloxacin            | 257 (59.8)    | 4 (0.9)          | 169 (39.3)     |
| Ofloxacin                | 231 (53.7)    | 13 (3.0)         | 186 (43.3)     |

PDR Pandrug-resistant, XDR Extensively drug-resistant, MDR Multidrug-resistant

tobramycin (32.6%), amikacin (27.4%), piperacillin (18.6%), piperacillin–tazobactam (17.6%), doripenem (17.0%), aztreonam (16.7%), meropenem (16.5%) and imipenem (15.8%).

Resistance of isolates was high to cefepime (65.6%), ofloxacin (62.2%), levofloxacin (56.7%), ciprofloxacin (56.7%), gentamicin (56.7%), tobramycin (51.1%), amikacin (42.2%), piperacillin (27.8%) and imipenem (27.8%) in UCHI. Of the seven hospitals, OTHI isolates showed highest resistance to ticarcillin (84.4%), cefoperazone (66.7%), ceftazidime (60%), piperacillin–tazobactam (31.1%), meropenem (28.9%), and aztreonam (28.9%). Isolates from FMCO had the least resistance to ciprofloxacin (28.9%) and gentamicin (35.6%), while isolates from FMCA had least resistance to tobramycin (13.3%), amikacin (11.1%), imipenem (2.2%), ticarcillin (48.9%), piperacillin (8.9%) and piperacillin–tazobactam (4.4%). Among the isolates, 56 (13.0%) exhibit resistance to imipenem, meropenem and doripenem, 9 (2.1%) exhibit resistance to both meropenem and doripenem, 7 (1.6%) exhibit resistance to both imipenem and doripenem, while 4 (0.9%) exhibit resistance to both imipenem and meropenem.

According to the criteria for classification of multidrug-resistant *P. aeruginosa* by [21], 109 (25.4%) 47 (10.9%) and 25 (5.98%) *P. aeruginosa* isolates were multidrug-resistant (MDR), extensive drug-resistant (XDR) and pandrug-resistant (PDR), respectively (Table 2). Fifty eight (58) different pattern of resistance was obtained to 181 isolates that were resistant to multiple antibiotics (Supplementary table 4). Multidrug resistance was highest among isolates from UCHI 33/90 (36.7%) followed OTHI 16/45 (35.6%), FMCI 31/90 (34.4%), FMCA 15/45 (33.3%), FMCO 25/90 (27.8%), OTHS 12/45 (26.7%) and LTHO 10/45 (22.2%). The XDR rates among the isolates in descending order: UCHI 17 (18.9%), OTHI 8 (17.8%), LTHO 7 (15.6%), FMCI 5 (7.1%), FMCO 6 (6.7%), OTHS 3 (6.7%), and FMCA 1 (2.2%). Combined disk test (CDT) phenotypically identified 73 (17.0%) of 430 isolates as MBL producers.

**Table 2** Antibiotic Profile Classification of clinical isolates of *P. aeruginosa* from Southwest Nigeria

| State          | Classification by drug resistance n = 430 (100%) |           |            |            |
|----------------|--------------------------------------------------|-----------|------------|------------|
|                | PDR n (%)                                        | XDR n (%) | MDR n (%)  | NMDR n (%) |
|                | 25 (5.8)                                         | 47 (10.9) | 109 (25.4) | 249 (57.9) |
| Oyo (n = 90)   | 8 (8.9)                                          | 19 (21.1) | 24 (26.7)  | 39 (43.3)  |
| Ogun (n = 90)  | 1 (1.1)                                          | 3 (3.3)   | 25 (27.8)  | 61 (67.8)  |
| Osun (n = 90)  | 9 (10.0)                                         | 11 (12.2) | 19 (21.1)  | 51 (56.7)  |
| Ondo (n = 90)  | 3 (3.3)                                          | 12 (13.2) | 17 (18.9)  | 58 (64.4)  |
| Ekiti (n = 70) | 4 (5.7)                                          | 2 (2.9)   | 24 (34.3)  | 40 (57.1)  |

## Minimum Inhibitory Concentration

Minimum inhibitory concentration range for meropenem was between 0.5 and  $\geq 64$   $\mu\text{g/mL}$ . Out of 430 isolates, 352 (81.9%) had MIC value of  $\leq 2$   $\mu\text{g/mL}$  (sensitive MIC value). Seventy-six (76) out of 181 isolates that were resistant to three or more classes of antibiotics by disk diffusion method gave MIC value of between 8 and  $\geq 64$   $\mu\text{g/mL}$  (resistant MIC value).

## Percentage Distribution of Carbapenemase Genes in Carbapenem-Resistant *P. aeruginosa*

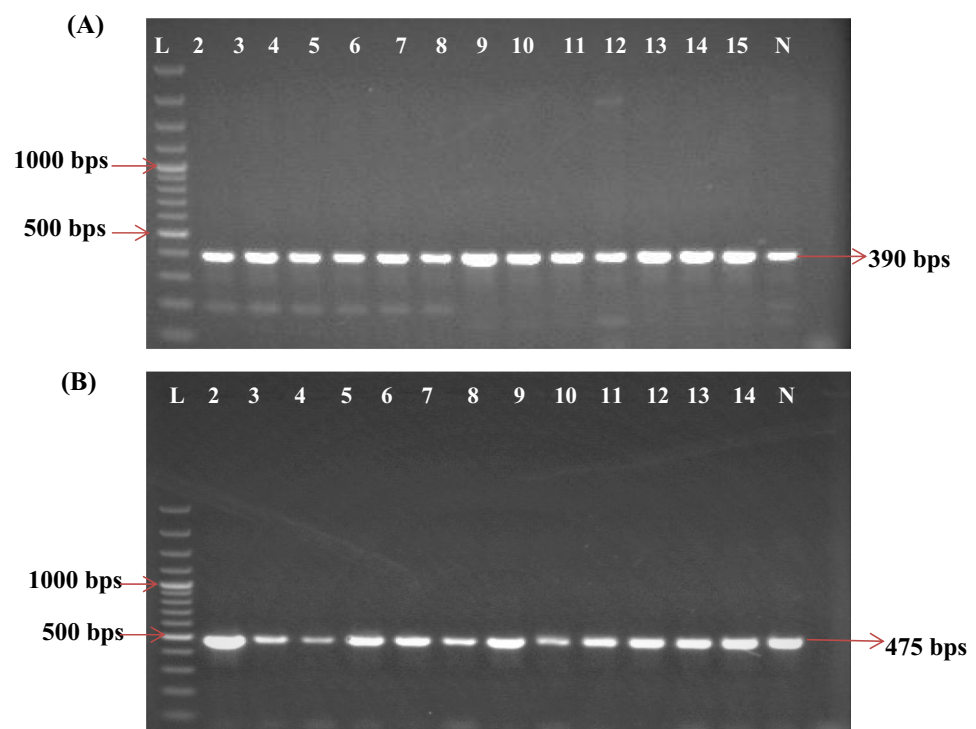
The result indicated that all the 73 carbapenem-resistant *P. aeruginosa* clinical isolates do not harbor *bla*<sub>SME</sub>, *bla*<sub>GES</sub>, *bla*<sub>NMC-A</sub>, *bla*<sub>BIC-1</sub>, *bla*<sub>OXA-48</sub> and *bla*<sub>OXA-58</sub>. Out of eight MBL-encoding genes, only two, *bla*<sub>VIM</sub> and *bla*<sub>NDM</sub> were seen in association with our isolates. The occurrence of these MBL genes in carbapenem-resistant *P. aeruginosa* was 86.3% (63/73). The *bla*<sub>VIM</sub> gene was detected in 26 (35.6%) carbapenem-resistant clinical isolates, while *bla*<sub>NDM</sub> was detected in 28 (38.4%) isolates. Singleplex PCR identified the presence of both *bla*<sub>VIM</sub> and *bla*<sub>NDM</sub> among 9 (12.3%) clinical isolates. Out of 28 isolates which harbored *bla*<sub>NDM</sub>, 14 were XDR, while 12 out of 26 isolates that harbored *bla*<sub>VIM</sub> alone were XDR. Twenty-one of 25 PDR isolates contained MBL gene(s), 12 had *bla*<sub>NDM</sub>, while 7 were *bla*<sub>VIM</sub> containing isolates. The *bla*<sub>VIM</sub> and *bla*<sub>NDM</sub> were each found in 21 (37.5%) of CRPA isolates

from wound, while 8 (14.3%) had both *bla*<sub>VIM</sub> and *bla*<sub>NDM</sub>. There was significant differences in resistance to the 16 antibiotics among strains encoding MBL genes and MBL-negative strains ( $p < 0.05$ ) (Supplementary table 5). The agarose gel electrophoresis of amplified *bla*<sub>VIM</sub> and *bla*<sub>NDM</sub> genes are shown in Figure 2.

Sequenced amplicons of nine *bla*<sub>VIM</sub> and nine *bla*<sub>NDM</sub> were submitted to Basic Local Alignment Search Tools (BLASTn) on National Center for Biotechnological Information (NCBI) website to determine their identity with BLASTN 2.9.0+ program. The best BLAST Hits with the submitted nucleotide sequence were chosen based on the maximum similarity with the sequence in the GenBank database. All the nine *P. aeruginosa* strains; PS183, PS152, PS335, PS184, PS367, PS154, PS303, PS395 and PS285 *bla*<sub>NDM-1</sub> sequences in this study showed 100% similarity with *bla*<sub>NDM</sub> sequence accession number MK371546 in the GenBank. The *bla*<sub>NDM</sub> sequences of the nine *P. aeruginosa* strains submitted to the GenBank were assigned the following accession numbers MN193051 - MN193059.

Six of the nine sequenced *bla*<sub>VIM</sub> amplicons of *P. aeruginosa* strains; PS220, PS209, PS210, PS243, PS168 and PS291 showed 99.73% similarity with *bla*<sub>VIM-5</sub> sequence accession number MK209000 in the GenBank, while the remaining three strains; PS219, PS204 and PS097 showed 99.48%, 99.46% and 99.23% similarity, respectively. The *bla*<sub>VIM</sub> sequences of these *P. aeruginosa* strains submitted to the GenBank were assigned the following accession numbers MN201592 - MN201600.

**Fig. 2** **A** Representative gel image of amplified *bla*<sub>VIM</sub> on 1.5% agarose gel *L* = 100 bp plus; Lane 2–15 shows different carbapenem non-susceptible *P. aeruginosa* positive for *bla*<sub>VIM</sub>; *N*: Negative control. **B** Representative gel image of amplified *bla*<sub>NDM</sub> on 1.5% agarose gel *L* = 100 bp plus ladder; Lane 2–14 shows different carbapenem non-susceptible *P. aeruginosa* isolates positive for *bla*<sub>NDM</sub>; *N*: Negative control



## Molecular Characterization of Carbapenem-Resistant *P. aeruginosa*

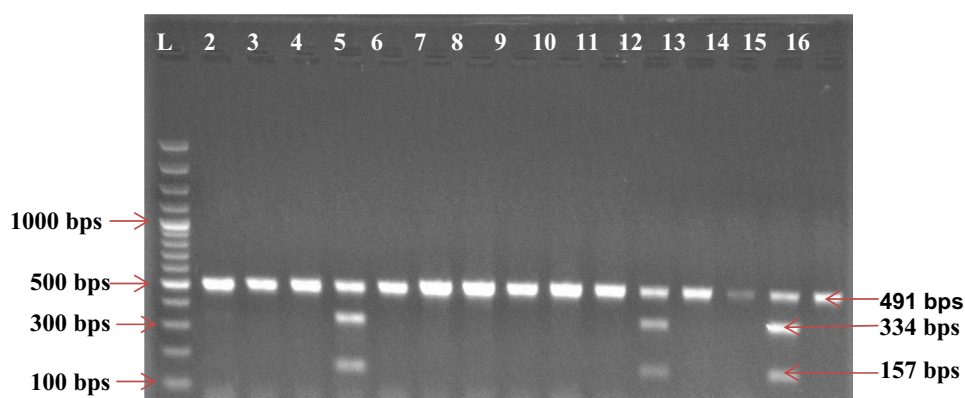
Figure 3 shows the agarose gel electrophoresis of integrons. Class 1 integrons (intI1 gene) were present in 42 (57.5%) of 73 carbapenem-resistant *P. aeruginosa* (CRPA), while classes 1 and 2 integrons were present in 9 (12.3%). None of the isolates possess single class 2 or class 3 integrons. Four of the ten (40%) carbapenem-resistant isolates that did not contain MBL gene harbor class I integron (intI1) gene. Of the 63 isolates carrying MBL genes, 45 (71.4%) harbor integrons. Class 1 integrons was most commonly detected in UCHI with 14 (69.6%) followed by FMCI and LTHO 5 (62.5%), and most commonly in isolates from wound 50 (89.3%). Among the 56 CRPA isolates from wounds, 27 (48.2%) harbor single class 1 integrons, while 20 (31.8%) lacked integrons. All the isolates that possessed both class 1 and 2 integrons 9 (16.1%) were from wound. Seven out of every 10 CRPA isolates from wound possessed integrons. Among CRPA isolates from urine samples, 6 (75%) out of 8 isolates harbor class 1 integrons. All CRPA from tracheal aspirate, ear and wound biopsy possessed single class 1 integrons. Four and seven isolates *bla*<sub>VIM</sub> and *bla*<sub>NDM</sub> genes, respectively, lacked integrons. Two isolates that harbored both *bla*<sub>VIM</sub> and *bla*<sub>NDM</sub> genes do not harbor integrons.

Thirty-four of the 51 class 1 integrons-containing CRPA isolates contained integrase gene cassettes. Amplified class 1 integrons gene cassette yielded 5 amplicon sizes approximately 250 bps to 4.0 kb; 17 (33.3%) gave a single fragment size of 4.0 kb, while 3 (5.9%) gave double fragment sizes of ~4.0 kb and 800 bp. Single fragment sizes of 1.6 kb, 1.4 kb, 800 bps and 250 bps were detected in 4 (7.8%), 1 (2.0%), 8 (15.7%) and 1 (2.0%) isolates, respectively. Eight of the 17 strains whose integrase could not be amplified were PDR,

while seven were XDR. Five of these strains also harbored class 2 integrons.

Upregulation of the four efflux systems (MexAB-OprM, MexCD-OprJ, MexEF-OprN and MexXY-OprM) was determined in carbapenem-resistant *P. aeruginosa* and one multidrug susceptible strain by determining the relative transcription levels of *mexA*, *mexB*, *mexC*, *mexD*, *mexE*, *mexF*, *mexX* and *mexY* genes by quantitative real-time PCR. The mRNA was said to be overexpressed if the equivalent mRNA level was at least 2-fold (*mexA*, and *mexB*), 4-fold (*mexX* and *mexY*) or 100-fold (*mexC*, *mexD*, *mexE* and *mexF*) higher than that for *P. aeruginosa* ATCC 27853. The gene expression results showed that 48 (65.8%) of the carbapenem-resistant *P. aeruginosa* clinical isolates highly expressed one or more efflux pump genes, while 25 (34.3%) isolates did not show any such trend. One isolate (PS293) have shown higher expression of maximum number of genes, i.e., *mexA*, *mexB*, *mexX*, *mexY*, *mexC*, *mexD* and *mexE*. MexXY-OprM was the significantly over expressed pump and was seen in 41 (56.2%) isolates followed by MexAB-OprM in 34 (46.6%) strains, MexCD-OprJ in 7 (9.6%) strains and MexEF-OprN in 6 (8.2%) of the isolates.

Among the carbapenem-resistant *P. aeruginosa* strains, *mexX* was the highly expressed gene in many of the isolates 39 (53.4%) studied, while *mexF* was the least expressed gene as reflected by its expression profile in association with only two strains (2.7%) (Table 3). In 26 (35.6%) isolates, both *mexX* and *mexY* genes were overexpressed, while *mexX* and *mexY* were overexpressed individually in 13 (17.8%) and 2 (2.7%) isolates, respectively. Both *mexA* and *mexB* genes were overexpressed in 20 (27.4%) of isolates, while *mexA* and *mexB* genes were overexpressed individually in (12.3%) and 6 (8.2%) isolates, respectively. Both *mexC* and *mexD* genes were highly expressed in 6 (8.2%) of the strains.



**Fig. 3** Agarose gel (1.5%) showing PCR products for integrons *L*=100 plus bp ladder; Lanes 2–4, 6–11, 13, 14, 16, 18 shows different carbapenem-resistant *P. aeruginosa* positive for class 1 integrons after digestion of PCR products with *RsaI* enzyme (PS154, PS181,

PS205, PS152, PS220, PS209, PS168, PS184, PS210, PS414, PS335, PS297); Lanes: 5, 12, 15 shows different carbapenem-resistant *P. aeruginosa* positive for class 1 and 2 integrons after digestion of PCR products with *RsaI* enzyme (PS183, PS285, 222)

**Table 3** MDR efflux gene overexpression, *ampC* overexpression and *oprD* loss in carbapenem-resistant *P. aeruginosa*

| Gene        | Gene expression fold range (No of isolates) |          |          |            |       | No (%) of isolates with overexpression or <i>oprD</i> loss |
|-------------|---------------------------------------------|----------|----------|------------|-------|------------------------------------------------------------|
|             | <0.01                                       | 0.01–5.0 | 5.1–10.0 | 10.1–100.0 | > 100 |                                                            |
| <i>mexA</i> | –                                           | 61       | 3        | 4          | 5     | 28 (38.4)                                                  |
| <i>mexB</i> | –                                           | 59       | 4        | 6          | 4     | 26 (35.6)                                                  |
| <i>mexC</i> | –                                           | 43       | 9        | 15         | 6     | 6 (8.2)                                                    |
| <i>mexD</i> | 1                                           | 44       | 10       | 11         | 7     | 7 (9.6)                                                    |
| <i>mexE</i> | –                                           | 62       | 1        | 6          | 4     | 4 (5.5)                                                    |
| <i>mexF</i> | –                                           | 61       | 4        | 6          | 2     | 2 (2.7)                                                    |
| <i>mexX</i> | –                                           | 39       | 6        | 21         | 7     | 39 (53.4)                                                  |
| <i>mexY</i> | –                                           | 47       | 4        | 16         | 6     | 28 (38.4)                                                  |
| <i>oprD</i> | 22                                          | 49       | 2        | –          | –     | 52 (71.2)                                                  |
| <i>ampC</i> | –                                           | 49       | 5        | 5          | 14    | 19 (26.0)                                                  |

mRNA expression level  $\geq$  twofold (*mexA*, *mexB*),  $\geq$  fourfold (*mexX* and *mexY*) or  $\geq$  100-fold (*mexC*, *mexD*, *mexE* and *mexF*)  $< 0.3$  *oprD*,  $\geq 50.0\%$  (*ampC*)

However, none of the isolates showed significant expression of both *mexE* and *mexF* genes.

The *ampC* gene was highly expressed in 19 (26.0%) strains with fold increase of between  $> 50.0$ . On contrary, the reduced expression of outer membrane porin (*oprD*) was observed in 48 (65.8%) carbapenem-resistant *P. aeruginosa* strains (Table 3). Among the ten carbapenem-resistant *P. aeruginosa* that lacked MBL genes, the only mode of resistance in four isolates was attributable to efflux pump overexpression (PS007, PS088, PS099, PS349), while defective *oprD* was independently responsible for carbapenem resistance in three isolates (PS150, PS166, PS185). Carbapenem resistance in PS170 and PS244 was due to cumulative higher expression of efflux pump and *ampC*, while in strain PS384, carbapenem resistance stems from the coordinated overexpression of efflux pump and under expression of *oprD*.

### Combination of Resistance Mechanisms in CRPA

In this study, majority of carbapenem-resistant strains possess more than one resistance gene (Supplementary table 6). Nine different combinations of resistance mechanisms were observed with combinations of MBL, Efflux pumps overexpression and *oprD* loss being the most prevalent (34.2%). The combinations of all the four resistance mechanisms (MBL, efflux pump overexpression, *oprD* deficient and *ampC* overexpression) were exhibited in 9 (12.3%) strains, while MBL and efflux pump alone was found in 11 (15.1%) strains. Combination of efflux pumps and *ampC* overexpression was detected in 6 (8.2%) strains, while MBL, *oprD* and *ampC* were present simultaneously in 2 (2.7%) strain only (Table 4). Among the ten isolates that lack MBL resistance genes, four demonstrated efflux pump overexpression alone, three isolates showed *oprD* loss alone while the remaining three isolates had the combination of efflux pump and *ampC* overexpression. The occurrence of MBL, class 1 integron,

**Table 4** Individual and combination of resistance mechanisms in carbapenem-resistant *P. aeruginosa* from Southwest Nigeria

| Mechanisms                                       | Number (%) |
|--------------------------------------------------|------------|
| MBL, Efflux pump, <i>oprD</i> deficient and AmpC | 9 (12.3)   |
| MBL, Efflux pump and <i>oprD</i> deficient       | 25 (34.2)  |
| MBL, Efflux pump and AmpC                        | 3 (4.1)    |
| MBL, <i>oprD</i> deficient and AmpC              | 2 (2.7)    |
| MBL and Efflux pump                              | 11 (15.1)  |
| MBL and <i>oprD</i> deficient                    | 9 (12.3)   |
| Efflux pump and AmpC                             | 3 (4.1)    |
| <i>oprD</i> deficient and AmpC                   | 1 (1.4)    |
| MBL and AmpC                                     | 1 (1.4)    |
| Efflux pump                                      | 4 (5.5)    |
| MBL                                              | 3 (4.1)    |
| <i>oprD</i> deficient                            | 2 (2.7)    |

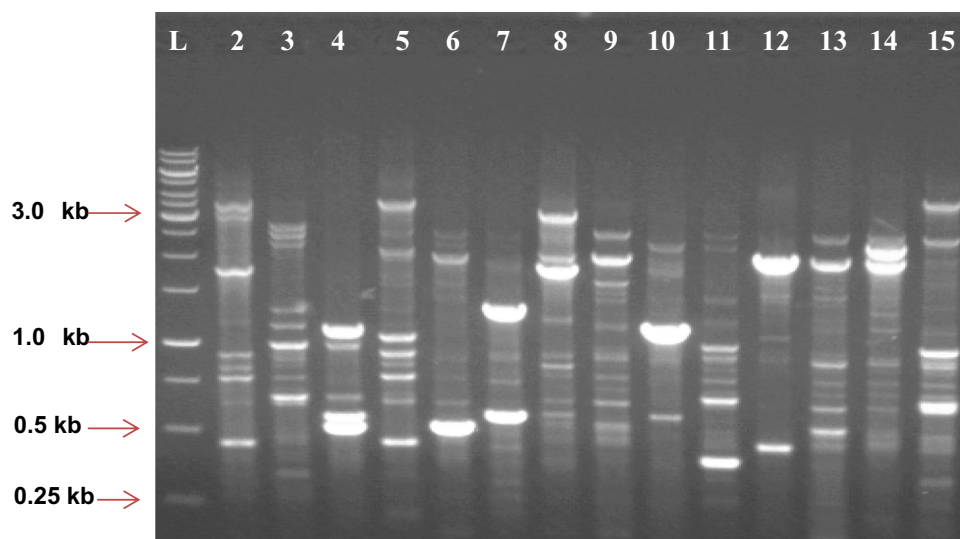
efflux pump, *oprD* and *ampC* genes in clinical isolates of *P. aeruginosa* from wound and non-wound sources was not significant ( $p > 0.05$ ) (Supplementary table 7).

### ERIC-PCR Profiles of Carbapenem-Resistant *P. aeruginosa*

Figure 4 shows the representative gel image of ERIC profile of carbapenem-resistant *P. aeruginosa*. Dice coefficient from ERIC pattern of each isolate was used to construct a dendrogram using UPGMA. Figure 5 represents the ERIC profiles of 73 carbapenem-resistant *P. aeruginosa* strains consisting of 14 MDR, 34 XDR and 25 PDR. The dendrogram revealed seven distinct clusters (I–VII) with cluster III being the predominant type containing 21 isolates followed by cluster V containing 15 isolates. The isolates exhibit high genetic heterogeneity and occurred as single clone with isolates from different hospitals clustering together. Cluster I consists of



**Fig. 4** Representative image of fingerprinting patterns of carbapenem-resistant *P. aeruginosa* strains by gel electrophoresis following ERIC-PCR; Lanes L = 1 kb plus ladder, Lanes 2–15 = PS414, PS383, PS202, PS297, PS335, PS147, PS394, PS367, PS250, PS293, PS405, PS007, PS400, PS285 and PS398



only three isolates from three different hospitals and cluster II comprises of 14 isolates from six hospitals. Cluster III being the predominant cluster composed of isolates from all the seven hospitals. All isolates in clusters I, IV and VI comprising of three, eight and four isolates, respectively, were XDR and PDR, while the remaining four clusters had MDR, XDR and PDR isolates. Majority of the isolates in cluster II were XDR, while cluster III had MDR, XDR and PDR isolates in equal number. Isolates which co-harbored *bla<sub>VIM</sub>* and *bla<sub>NDM</sub>* genes were associated with clusters I, II, IV and VII.

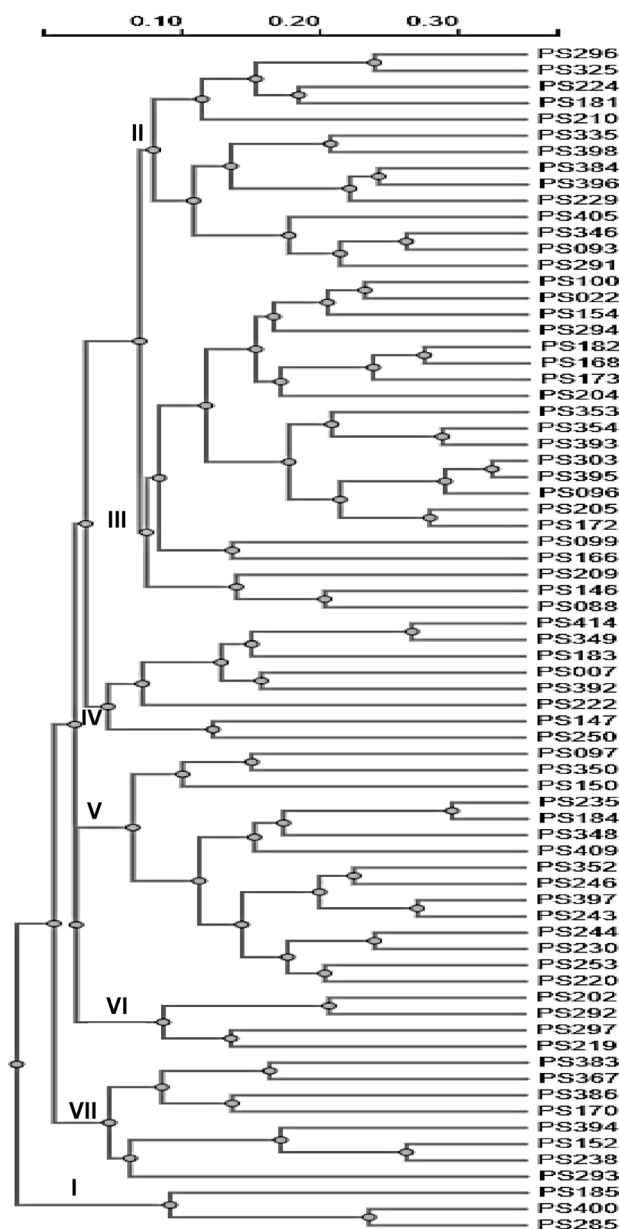
## Discussion

The ability of *P. aeruginosa* to evade the bactericidal effect of many antibiotics makes its control challenging [3]. All the *P. aeruginosa* isolates tested in this study were resistant to four or more antibiotics. This present study reports higher resistance rate to third and fourth generation cephalosporins with percentage resistance of 45.8% and 38.1% to cefepime and ceftazidime, respectively, in contrast to the report from the same region who recorded respective resistance rate of 9.3% and 14.8% [28]. Higher resistance to cefepime was observed in this study when compared with previous data from the same southwest region where 6.0% were resistant [28]. High level of resistance to most of antibiotics tested during this study could be linked with unnecessary use of expanded spectrum antibiotics and hence strong antibiotic stewardship should be incorporated in essential health policies. Increasing resistance to cephalosporins, most especially the fourth generation cephalosporin, necessitates the need for proper guideline of antibiotic prescriptions. In this study, only 21.9% CRPA were susceptible to aztreonam which is a beta-lactam with efficacy against MBL-producing

Gram-negative bacteria. However, aztreonam could not resist hydrolysis by ESBLs suggesting the presence of ESBLs among the carbapenem-resistant *P. aeruginosa*.

Carbapenems are potent antibiotics that are used as last resort in the therapy of Gram-negative bacterial infections especially those due to ESBL-producing organisms. However, resistance to carbapenems is on the rise [29]. Carbapenem resistance as recorded in this study was defined as non-susceptibility to at least one anti-pseudomonal carbapenem. Overall prevalence of 18.8% CRPA reported in this study is in the range of 15.2% reported from Nigeria and 19.6% from India [30, 31]. In the systematic review on carbapenemase producing Gram-negative bacteria in hospital settings in Africa, overall prevalence ranged from as low as 2.3% and 9.0% to as high as 60.0% and 67.7% in sub-Saharan Africa and North Africa, respectively [32]. Among the carbapenems, reduced resistance to imipenem (15.8%) was observed compared to meropenem (16.5%) and doripenem (17.0%) in this study. All CRPA isolates (100%) in this study were resistant to three or classes of antibiotics, which was higher than 74.4% reported by Palavecino and Kilic [33]. In a previous study on multidrug-resistant *P. aeruginosa* from southwest Nigeria, [28] reported the prevalence of 1.9% resistance to imipenem, much lower than the 15.8% recorded in our study. Similarly, resistance to doripenem was high in this study. This is because doripenem is structurally related to imipenem leading to cross-resistance to this antibiotic. This also calls for concern as this antibiotic is not in clinical use in Nigeria.

Genes encoding class A (*bla<sub>SME</sub>*, *bla<sub>NMC-A</sub>*, *bla<sub>GES</sub>* and *bla<sub>BIC-1</sub>*) and class D (*bla<sub>OXA-48</sub>* and *bla<sub>OXA-58</sub>*) carbapenemases were not detected among the *P. aeruginosa* isolates in this study. Although there are several reports of *bla<sub>OXA-48</sub>* in Gram-negative bacteria including *P. aeruginosa* from African countries, *bla<sub>OXA-58</sub>* is not common



**Fig. 5** Dendrogram showing cluster analysis of carbapenem-resistant *P. aeruginosa* strains by ERIC-PCR using Phylotree software and Unweighted Pair Group Method with Arithmetic Mean (UPGMA)

[31]. However, blaGES has been reported in *P. aeruginosa* from various countries [34, 35]. The overall prevalence of MBL genes (14.7%) in *P. aeruginosa* in our study in South-west Nigeria was higher than the rate of 2.5% reported from North Central Nigeria [13]. The prevalence of blaVIM and blaNDM in CRPA in this study was 86.3% (63/73) significantly higher than 5% and 55% documented for similar genes in CRPA from Turkey and India, respectively [30, 36]. Previous study from Africa have also reported the presence of blaVIM-2 in *P. aeruginosa* [37], which has been confined to *P. aeruginosa* with the exception of the report of blaVIM-2

producing *E. coli* from Tunisia [32]. Previous study reported the presence of blaVIM-2 in clinical isolates of *P. aeruginosa* from Nigeria [13], but at present there is no report of blaVIM-5 in clinical isolates of *P. aeruginosa* from Africa, although a recent study reported blaVIM-5 in environmental strains of *Pseudomonas putida* from Nigeria [38]. Our study is the first report of blaVIM-5-like in clinical isolates of *P. aeruginosa* from Nigeria

In Africa, blaNDM-1 was first detected in *Klebsiella pneumoniae* isolated from urine sample of a patient in Kenya [39], but there are limited data with respect to blaNDM-1 in Gram-negative bacteria from Nigeria. The presence of blaNDM-1 in Enterobacteriaceae from Nigeria has been reported [40]. Recent study reported the absence of blaNDM-1 in clinical isolates of *P. aeruginosa* [13]. To the best of our knowledge, this is the first report of blaNDM-1, and also the first report of co-existence of blaNDM-1 with blaVIM-5-like in clinical isolates of *P. aeruginosa* from Nigeria. The co-existence of blaVIM with blaNDM in clinical isolates of *P. aeruginosa* from India and Pakistan has been documented [41, 42]. Our study did not detect blaIMP, blaSIM, blaSPM, blaGIM, blaAIM and blaDIM and there have been no reported cases of these MBL-encoding genes from Nigeria.

The role of integrons in the dissemination of antibiotic resistance genes in Gram-negative bacteria is well documented [32, 43, 44]. Single class 1 integrons was detected in 57.5% of carbapenem-resistant *P. aeruginosa* in our study which agrees with the report from the same region with prevalence of 57% in clinical isolates of *P. aeruginosa* [43], but lower than the report from Egypt with 71% prevalence [45]. Moreover, this is the first report of co-existence of class 1 as well as class 2 integrons in *P. aeruginosa* from Nigeria. In this study, two strains with co-occurrence of blaVIM-5-like and blaNDM-1 genes did not show presence of integrons, which agrees with the report of [13] where all their blaVIM-2 isolates also did not harbor integrons. This study was able to amplify cassette region that was not greater than 4.0 kb in size but gene cassette could not be amplified in 17 (33.3%) integrons-containing strains. This means that these strains harbored integrons with molecular weight of >4.0kb. Since 15 of these 17 strains were XDR and PDR, the implication is that class 1 integrons present in these strains have acquired multiple-resistant genes.

MexXY-OprM was the most expressed efflux pump in this study with 56.2% prevalence. This is in support of the work from Brazil with MexXY-OprM being the most expressed pump with prevalence of 50.8% in *P. aeruginosa* [46]. MexAB-OprM was overexpressed in 46.6% of isolates in this study. This was lower than the report from Iran which showed that 62% of *P. aeruginosa* overexpressed MexAB-OprM [47]. Both MexAB-OprM and MexXY-OprM was overexpressed simultaneously in 23(31.5%) isolates.

Co-expression of two or more efflux pumps in clinical strains of *P. aeruginosa* has been documented. For example, concomitant overexpression of MexAB-OprM, and MexXY-OprM showing resistance to multiple antibiotics in *P. aeruginosa* was reported from France [48]. MexAB-OprM and MexXY-OprM were naturally expressed in 'wild type' cell. MexCD-OprJ and MexEF-OprN were not active in unmutated cell but overexpressed in mutants [15]. Since all these carbapenem-resistant strains were also impervious to many other classes of antibiotics together with fluoroquinolones, aminoglycosides, beta-lactams and monobactam, which are substrates for these efflux pumps, this underscores the task of efflux pumps in antibiotic resistance.

The current study also showed that 65.8% of CRPA had oprD downregulation. OprD porin is the only porin through which carbapenems enter the bacterial cell. Mutation or alteration in genes that code for OprD protein results in resistance to carbapenems and reduced expression of oprD. Deficient OprD porin is mainly seen in carbapenem-resistant *P. aeruginosa* [48]. Deficient oprD as a result of mutations has been reported to confer high resistance to imipenem than meropenem and doripenem [49]. Carbapenem resistance in three strains was due to loss of OprD porin alone. MBL-encoding genes, efflux pumps and ampC overexpression were not seen in these isolates suggesting that deficient OprD porin alone could contribute to carbapenem resistance.

Similarly, increased ampC expression was found in 19 (26.0%) isolates. This outcome was in concordance with the report of [8] who stated that 24.2% of the isolates overexpressed ampC but higher than [46] who documented that 11.9% of *P. aeruginosa* isolates from Brazil overexpressed ampC. The overexpression of ampC as a sole resistance mechanism could not be linked with carbapenem resistance in this study, because all the isolates which overexpressed this gene also have other carbapenem-resistant mechanisms. This was backed up with the report that ampC overexpression alone in *P. aeruginosa* could only lead to insignificant reduction in the activity of carbapenems but combination of other mechanisms of resistance could have a great impact in lowering the activity of carbapenems [49]. Therefore, ampC overexpression alone may not necessarily be responsible for carbapenem resistance. This study for the first time in Nigeria has quantified gene expression (efflux pumps, oprD and ampC) in *P. aeruginosa*. This study also identified the entire four clinically important efflux pumps in *P. aeruginosa*.

The two MDR strains from current study having intact OprD porin and lacked carbapenemase genes seemed to highly express MexAB-OprM, MexXY-OprM, MexEF-OprN and ampC suggesting the impact of efflux pumps and ampC in antibiotic resistance of these isolates. Co-expression of MexAB-OprM, MexXY-OprM, MexCD-OprJ, ampC with deficient oprD was also observed in two strains. This shows interconnection of various mechanisms of resistance

with a significant effect on resistance to carbapenem. Gomaa et al [50] documented the presence of all the four efflux pumps (MexAB-OprM, MexXY-OprM, MexCD-OprJ and MexEF-OprN) in 15% of *P. aeruginosa*; however, PS293 (2.1%) isolate was found to be overexpressing all the four efflux pumps in this study. This isolate was deficient in oprD showing, -increased expression of ampC and possessed all the four efflux pumps suggesting multiple mechanisms of resistance. This isolate was also resistant to all the antipseudomonal antibiotics used in this study. Moreover, ampC overexpression alone was not detected as the only mechanism of resistance to carbapenems.

ERIC-PCR for determining genetic relatedness or clusters has been successfully employed by various researchers for typing *P. aeruginosa* [42, 51]. ERIC-PCR dendrogram of carbapenem-resistant isolates showed diversity or heterogeneity vis a vis their source of isolation from the same hospital as well as from different hospitals indicating genetically or wide clonal distribution of MDR isolates. Interestingly, there was also clonal relatedness among small number of carbapenem-resistant isolates from different hospitals. This clonal relatedness was reflected by a pair of XDR isolates from wound that come from different hospitals exhibit. These isolates harbored bla<sub>NDM</sub> and class 1 integrons. This observation showed that there is limited transmission of resistant isolates in the seven tertiary hospitals. However, the presence of MBL resistance genes, together with highly expressed efflux pumps and other resistance genes encoded in these isolates is of major concern in the hospital settings. In nutshell, AMR is a serious challenge and poor management of regulated use of antibiotics, which due to individuals irrational behavior, acting without any check and balance, can cause long-term harm to public health.

## Conclusions

This study reports high incidence of presence of transmissible MBL genes, co-occurrence of bla<sub>VIM-5</sub>-like and bla<sub>NDM-1</sub> genes, and co-existence of class 1 and 2 integrons in carbapenem-resistant *P. aeruginosa* isolates from Nigeria. Furthermore, the hyperactive/highly expressed efflux pumps, i.e., MexAB-OprM, MexCD-OprJ, MexEF-OprN and MexXY-OprM as well as oprD and ampC in *P. aeruginosa* isolates suggest the altered drug efflux further resulting in emergence of multidrug resistance. ERIC-PCR revealed a great level of genetic diversity among carbapenem-resistant *Pseudomonas aeruginosa* strains suggesting that any future outbreak due to such highly resistant cluster of *Pseudomonas aeruginosa* may ensue. This is a serious hazard to health care system and underscores the need for regular surveillance and monitoring of resistance to carbapenems and associated resistance genes which is usually used for empirical treatment of high risk

patients. The presence of different combinations of efflux pumps and their variable expression, widespread association of MBLs with concomitant deficiency of *oprD* and overexpressed *ampC* all contribute to resistance mechanisms and hence effective antibiotic stewardship is required reduce the emergence of highly antibiotic-resistant isolates.

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**Authors Contributions** OEA and HB conceived and designed the study; OEA, HB and RS coordinated the experiment; OBO carried out the experiments, drafted the manuscript and analyzed the data. AD, HB did necessary editing of the manuscript. All authors read and approved the manuscript.

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**Data Availability** Data available within the article or its supplementary materials

**Code Availability** Not applicable.

## Declarations

**Conflict of interest** The authors report no conflicts of interest in this work.

**Ethical Approval** The study was approved in Ethical Committee of Department of Biosciences, COMSATS University, and Islamabad in joint study group (with Approval No. 3240293214). Isolates were taken from the Hospital laboratories in Southwest Nigeria.

**Consent to Participate** Not applicable.

**Consent for Publication** Not applicable.

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